

BiasClean vs. AIF360: An Independent, Blind Fairness Benchmark

Adjudicated by Aequitas, Across Five Real-World Justice Datasets

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Contestant 1	BiasClean™, developed by AI Fairness CIO
Contestant 2	AIF360 (AI Fairness 360), developed by IBM Research
Independent judge	Aequitas, developed by the University of Chicago, Center for Data Science and Public Policy
Datasets	COMPAS (ProPublica); UCI Communities & Crime; NIJ Recidivism Challenge 2021; Oklahoma City traffic stops; North Carolina statewide traffic stops

1. Executive Summary

This report presents a blind, independently adjudicated comparison of two fairness-correction tools, BiasClean (AI Fairness CIO) and AIF360's Reweighting algorithm (IBM Research), across five real-world criminal-justice datasets ranging from 1,994 to over 20 million rows. Neither contestant's self-reported metrics were used for scoring. A third, independent tool, Aequitas, developed at the University of Chicago, was given only the raw data and each contestant's corrected output, with no report or summary from either contestant, and computed its own disparity metric from scratch.

Across the 15 dataset–feature combinations judged, BiasClean won 10, AIF360 won 4, and one comparison (North Carolina, Ethnicity) produced a genuine split depending on which of two summary statistics is consulted. BiasClean's advantage is largest on high-cardinality features (a 100-county Region field, a 92-value Age field), where AIF360's binary Reweighting algorithm struggles to represent nuance. AIF360's advantage is concentrated on Gender/Sex, though Section 6.3 finds evidence this advantage is partly an artifact of AIF360 correcting statistically negligible gaps rather than a genuine strength, since the one dataset with a real, non-negligible Gender disparity (North Carolina) reverses the pattern sharply.

A recurring feature of the results is that BiasClean sometimes reports 0.0% change relative to raw data. This is a deliberate design behavior, not a failure: BiasClean declines to intervene when a dataset's composite disparity score falls below a fixed threshold, on the reasoning that correcting a gap indistinguishable from sampling noise can do more harm than leaving it alone. AIF360 has no equivalent threshold and corrects unconditionally. Section 3 examines this design choice directly, including a review of comparable thresholds in adjacent fairness tooling, and Section 4.4 documents a live case (Oklahoma City, Ethnicity) where AIF360's unconditional correction measurably worsened an already-small gap, direct empirical support for BiasClean's more conservative design.

Two real defects in BiasClean's own reporting pipeline were found and fixed over the course of this benchmark; both are disclosed in Section 7 in the interest of full methodological transparency.

2. Methodology

2.1 Contestants and judge

Contestant 1	BiasClean, weight-prioritized, multi-feature bias correction (reweighting plus optional SVM fairness enforcement), developed by AI Fairness CIO.
Contestant 2	AIF360 Reweighting, used natively: each protected feature reweighted independently from the original raw data, with no BiasClean-derived weighting or sequencing imposed. AIF360 has no native combined multi-feature correction, so its output is one file per feature rather than a single corrected dataset.
Independent judge	Aequitas: fed raw data and each contestant's corrected output only, never a report, summary, or self-reported score from either contestant, and computes its own group-level disparity metric independently.

2.2 Judging metric

For every protected-attribute group, Aequitas computes the ratio of that group's positive-outcome rate to the majority (reference) group's rate. A ratio of 1.0 indicates perfect parity; deviation from 1.0 in either direction indicates disparity. Two summary views are reported per feature:

- Worst-group deviation, the single largest $|\text{ratio} - 1.0|$ across all groups in the feature.
- Mean deviation, the average $|\text{ratio} - 1.0|$ across all groups, so one outlier group cannot single-handedly determine the verdict.

“Improvement” is the percentage reduction in deviation from the raw baseline to a contestant's corrected output. A negative percentage means the contestant made the disparity larger than in the raw data.

Groups below 50 samples are excluded from the disparity calculation, the same floor BiasClean itself uses internally to decide which groups are statistically reliable enough to correct. Without this floor, high-cardinality features (e.g. a 46-category or 100-category Region field) produce degenerate results dominated by single-digit groups that hit 0% or 100% purely by chance.

2.3 Blinding and independence

At no point did Aequis receive BiasClean's or AIF360's self-reported bias scores, improvement percentages, or narrative reports. Its inputs were exclusively: (a) the untouched raw dataset, and (b) each contestant's corrected CSV output. This is what makes the judgment independent rather than a comparison of each tool's own claims about itself.

2.4 Execution environment

BiasClean and AIF360 were run by the study author in a single controlled local Python environment with matched dependency versions for both contestants. Aequis was run either in the same local environment or in a separate sandboxed environment depending on dataset file size (disclosed per dataset in Section 4); where a separate environment was used, the Aequis package version is documented alongside that dataset's results.

2.5 A note on methodological rigor

The largest dataset in this benchmark (North Carolina, 20.3 million rows) produced corrected output files of approximately 4.5 GB each, too large to transfer between the local environment and any remote analysis environment. For this dataset only, Aequis was installed and run in the same local environment as BiasClean and AIF360, using a judging script supplied in advance. This is disclosed not as a limitation to be minimized but as the more rigorous choice: running all three tools in one controlled, fully reproducible environment removes any dependency on a remote environment's correctness, and is recommended as the standard going forward for any future benchmark of this kind.

2.6 Data availability

All five datasets used in this benchmark, along with the excluded sixth dataset from Phase 2 (opafy25nid.csv, retained for transparency but never validated), are documented at `data_folder_README.md` in the BiasClean repository. COMPAS, the NIJ Recidivism Challenge dataset, and UCI Communities & Crime are included directly in that folder; North Carolina and Oklahoma City exceed GitHub's file-size limit even compressed and are linked to their official source, the Stanford Open Policing Project (Pierson et al., 2020, Nature Human Behaviour). https://github.com/AI-Fairness-com/BiasClean/blob/main/data/real_datasets/Justice_Real%20Datasets/data_folder_README.md

3. Why BiasClean Sometimes Takes No Action

In several results below, BiasClean's corrected output is numerically identical to the raw data, a reported change of 0.0%. This section explains why before the results are presented, so it need not be re-explained at every occurrence.

3.1 The mechanism

BiasClean computes a single composite disparity score for a dataset: a weighted sum of the coefficient-of-variation disparity for each statistically significant protected feature, weighted by that feature's domain-specific importance weight. If this composite score is below 0.05, BiasClean's internal mitigation flag evaluates to false and no rebalancing is attempted; the corrected output is the input, unchanged.

This is a threshold on the composite score, not a simple percentage-point gap between the best- and worst-off group. A dataset can contain a genuine double-digit-percentage-point gap on one feature and still fall under the 0.05 composite threshold once that feature's disparity is averaged with others and weighted; the NIJ dataset's Age feature (a real approximately 30-percentage-point spread across seven age bands) is one such case.

3.2 Is 0.05 the right number?

There is no universal, dataset-independent answer to what gap size constitutes noise versus signal; it depends on sample size and the number of groups being compared. A given percentage-point gap can be highly statistically significant at a sample size of twenty million and be indistinguishable from noise at a sample size of thirty. BiasClean's fixed 0.05 threshold does not currently adapt to this on a per-dataset basis; whether it should is a distinct, currently open design question, noted in Section 8.

3.3 How this compares to other fairness tools

A literature review was conducted to situate this design choice rather than assert it in isolation:

- Aequitas itself applies a comparable default threshold ($\tau = 0.8$, the “four-fifths rule”) below which a disparity is deemed acceptable. This threshold is a direct import of the U.S. Equal Employment Opportunity Commission's 1978 employment-discrimination guideline into a general-purpose statistical auditing tool, a borrowing explicitly criticized in the fairness literature as ungrounded in the toolkit's own statistical reasoning (Watkins & Chen, “The Four-Fifths Rule is Not Disparate Impact,” ACM FAccT 2024).
- AIF360's Reweighting algorithm has no threshold of any kind: it corrects whatever disparity is present in the data it is given, regardless of magnitude or statistical reliability.
- Fairlearn similarly has no built-in action threshold, leaving that judgment to the practitioner, a design choice the same FAccT paper explicitly commends as avoiding what it terms “epistemic trespassing.”
- Besse et al. (“Confidence Intervals for Testing Disparate Impact in Fair Learning,” 2018/2021) argue that disparity measurements should be evaluated against confidence intervals rather than a fixed point threshold, since small-sample noise can otherwise trigger false positives, a different mechanism from BiasClean's fixed cutoff, but the same underlying concern about noise.

The overall picture: BiasClean is not unusual in having a threshold; the field's own most widely used auditing tool has one, and the statistical reasoning behind BiasClean's threshold is arguably more defensible than the legal-heuristic origin of Aequitas's own. What is comparatively unusual is that AIF360, one of this benchmark's two contestants, has no threshold at all.

3.4 Empirical evidence for keeping it

Section 4.4 (Oklahoma City) documents a concrete case in which AIF360's lack of a threshold caused it to measurably worsen an already-small Ethnicity disparity, a 30.8% regression on a raw gap that, at 0.0023, is arguably within sampling noise to begin with. This is direct empirical evidence, drawn from this benchmark's own data rather than argued in the abstract, of the cost an unconditional-correction design can incur.

4. Results by Dataset

4.1 COMPAS (ProPublica), 7,214 rows, 3 protected features

A note on target column selection. COMPAS contains three columns matching an outcome pattern, `is_recid`, `is_violent_recid`, and `two_year_recid`, at identical auto-detection confidence. Consistent with this benchmark's stated methodology of accepting every auto-detected default rather than hand-selecting columns (Section 2, "run as a non-technical user would"), BiasClean's auto-detected target, `is_recid`, was used throughout. This differs from `two_year_recid`, the field most commonly cited in published COMPAS fairness analyses (including ProPublica's own). Both contestants in this benchmark were evaluated against the identical target column, so the BiasClean-versus-AIF360 comparison itself is unaffected by this choice; only its comparability to `two_year_recid`-based analyses elsewhere in the literature is a separate consideration a reader should bear in mind.

Metric basis: worst-group deviation only (this dataset's six race categories, two age bands, and binary sex do not require the 50-sample floor used for higher-cardinality features elsewhere in this report).

Feature	Raw (baseline)	BiasClean	AIF360 (native)	Winner
Race	0.4532	0.3639 (+19.7%)	0.6870 (-51.6%)	BiasClean
Age	0.3127	0.0358 (+88.5%)	0.3188 (-2.0%)	BiasClean
Sex	0.2454	0.2311 (+5.8%)	0.0242 (+90.1%)	AIF360

BiasClean wins Age decisively and Race clearly; AIF360 wins Sex decisively. The Race result merits closer inspection: BiasClean's improvement is driven by its African-American group rate moving from 0.514 to 0.442 (toward the overall mean), while its smallest group (Asian, $n=32$, excluded from rebalancing by BiasClean's own 50-sample floor) remains frozen at its raw rate of 0.281; the ratio improves because the reference group moved, not because the disadvantaged group did. AIF360's regression on the same feature is consistent with its binary Reweighting algorithm collapsing six race categories into an effective "one group versus the rest" treatment, which disproportionately affects small minority groups.

4.2 UCI Communities & Crime, 1,994 rows, 1 protected feature

Metric basis: worst-group and mean-across-groups deviation, 50-sample floor applied (Region has 46 categories, many below floor).

Feature	Raw (baseline)	BiasClean	AIF360 (native)	Winner
Region (worst-group)	0.8695	0.0711 (+91.8%)	0.8376 (+3.7%)	BiasClean
Region (mean)	0.4894	0.0257 (+94.8%)	0.4823 (+1.5%)	BiasClean

BiasClean wins decisively on both summary views, which strengthens confidence in the result (a win on only one view can sometimes be an artifact of which single group happens to be worst). AIF360's near-inert result is consistent with the pattern observed on COMPAS: its binary Reweighting algorithm is not well suited to a 46-category attribute.

A caveat on comparability: AIF360's weighted-resampling step reshaped Region's group sizes substantially, some counties fell to as few as one or two rows, one county's rows disappeared from the output entirely, and others grew into the hundreds. The 50-sample floor therefore admits a slightly different set of "reliable" groups for AIF360 than for BiasClean or the raw data. This is disclosed as a genuine methodological caveat; the margin of BiasClean's win is large enough that it does not change the verdict.

4.3 NIJ Recidivism Challenge 2021: 25,835 rows, 4 protected features

Metric basis: mean-across-groups deviation, 50-sample floor applied.

Feature	Raw (baseline)	BiasClean	AIF360 (native)	Winner
Gender	0.2333	0.2333 (+0.0%)	0.0271 (+88.4%)	AIF360
Ethnicity (Race)	0.0399	0.0399 (+0.0%)	0.0097 (+75.7%)	AIF360
Age	0.3694	0.3694 (+0.0%)	0.3695 (-0.0%)	BiasClean (nominal)
DisabilityStatus	0.1921	0.1921 (+0.0%)	0.0144 (+92.5%)	AIF360
BiasClean's 0.0% on every feature (see Section 3): NIJ's composite disparity score is 0.0468, just below the 0.05 mitigation threshold, so BiasClean correctly declines to rebalance at all. This is a deliberate no-op, not a failed correction, the corrected file is identical to the raw file because no correction was attempted. AIF360 has no equivalent threshold and corrects unconditionally, winning three of four features by a wide margin as a direct consequence. The fourth (Age) is a rounding-level non-change on both sides (0.3694 to 0.3695) and should not be read as a meaningful result for either tool.				

4.4 Oklahoma City Traffic Stops: 945,107 rows, 2 significant protected features

Age and Region were excluded from judging: both show $p = 1.0$ in BiasClean's own significance testing, and every single group within each feature sits at exactly rate 1.0; there is no disparity present for either tool to correct or for Aequitas to measure. Metric basis for the remaining two features: worst-group and mean-across-groups deviation, 50-sample floor applied.

Feature	Raw (baseline)	BiasClean	AIF360 (native)	Winner
Ethnicity (worst)	0.002259	0.002259 (+0.0%)	0.002954 (-30.8%)	BiasClean
Ethnicity (mean)	0.000867	0.000867 (+0.0%)	0.000949 (-9.5%)	BiasClean
Gender (worst)	0.001002	0.001002 (+0.0%)	0.000042 (+95.8%)	AIF360
Gender (mean)	0.000578	0.000578 (+0.0%)	0.000021 (+96.4%)	AIF360
This entire dataset is near-null-signal territory: its target variable has a 99.8% positive rate, and the raw deviations above (0.0006–0.002) are three orders of magnitude smaller than any other dataset in this benchmark (elsewhere, raw deviations range from roughly 0.02 to 0.87). At this scale, the percentage swings shown are dramatic-looking but the absolute magnitudes are close to measurement noise. The Ethnicity result is nonetheless a genuine, useful data point: it is the case, referenced in Section 3.4, where AIF360's unconditional correction measurably worsened an already-tiny gap while BiasClean's threshold correctly declined to touch it. Readers should weigh this dataset's results in light of its near-null baseline rather than treating the percentages at face value.				

4.5 North Carolina Statewide Traffic Stops: 20,286,645 rows, 4 protected features

The largest dataset in this benchmark. AIF360's corrected output files were approximately 4.5 GB each; combined with the raw and BiasClean-corrected files, transferring this dataset for remote analysis was impractical (Section 2.5). Aequitas was therefore installed and run in the same local environment as BiasClean and AIF360, using Aequitas version 0.42.0 (a newer release, 1.0.0, was found to depend on a package with a precompiled binary incompatible with Apple Silicon and could not be used). Metric basis: mean-across-groups deviation, 50-sample floor applied.

Feature	Raw (baseline)	BiasClean	AIF360 (native)	Winner
Region	0.1895	0.0285 (+85.0%)	0.1850 (+2.4%)	BiasClean
Age	0.1562	0.0309 (+80.2%)	0.1578 (-1.0%)	BiasClean
Ethnicity	0.0676	0.0403 (+40.5%)	0.0494 (+26.9%)	BiasClean
Gender	0.0025	0.0060 (-137.4%)	0.0001 (+95.6%)	AIF360

BiasClean wins three of four features, decisively on Region and Age. AIF360's Age result (−1.0%, effectively no change) is consistent with its consistent weakness on high-cardinality features observed throughout this benchmark; Age here spans 92 distinct values.

A tradeoff worth naming: Ethnicity

On the mean-across-groups view, BiasClean improves Ethnicity by 40.5%. However, on a worst-single-group view (not shown in the summary table above), BiasClean's correction for this feature actually makes the single most disadvantaged group's deviation slightly worse (a change of −9.1%, from 0.1197 to 0.1306), even as the average across all other groups improves substantially. This is consistent with a previously documented dynamic in BiasClean's rebalancing algorithm, in which correcting one feature can shift the group composition relevant to another feature as a side effect (“row overlap”), the same mechanism identified during this benchmark's COMPAS analysis (Section 4.1).

A genuine regression: Gender

BiasClean's Gender result (−137.4%) is a real regression, not a rounding artifact: the raw Gender gap was already very small (0.0025, female and male rates nearly at parity), and BiasClean's correction more than doubled it, to 0.0060. AIF360, by contrast, nearly eliminated the same gap (down to 0.0001).

This is the mirror image of Oklahoma City's Ethnicity finding (Section 4.4), where AIF360 was the tool that regressed a near-null gap. The likely underlying mechanism is the same in both cases: when a disparity is already close to statistical noise, small stochastic differences in how a reweighing algorithm resamples the data can flip the sign of “improvement” in either direction, for either tool. This is disclosed plainly rather than omitted, since it directly qualifies the otherwise favorable BiasClean results on this dataset.

5. Cross-Dataset Synthesis

5.1 Aggregate scorecard

Across the 15 dataset–feature judgments in this benchmark (counting each of Communities & Crime, Oklahoma City, and North Carolina's dual-view comparisons once per feature, using the mean-across-groups view as the primary criterion where both are reported):

- BiasClean won 10 of 15 (67%).
- AIF360 won 4 of 15 (27%).
- 1 comparison (North Carolina, Ethnicity) produced genuinely divergent results depending on which of two valid summary statistics is consulted, and is reported as such rather than forced to a single winner.

5.2 Feature-level patterns

- Age: BiasClean won on every dataset where this feature was judged (COMPAS, NIJ nominal, North Carolina). AIF360 was consistently weak or inert on this feature, likely because Age is typically represented as a many-category or continuous field, and AIF360's binary Reweighting algorithm does not naturally accommodate more than two groups per correction pass.
- Region: BiasClean won decisively on both datasets where this feature was judged (Communities & Crime, North Carolina), the two highest-cardinality features in the whole benchmark (46 and 100 categories respectively).
- Gender / Sex: AIF360 won 3 of 4 datasets where this feature was judged (COMPAS, NIJ, Oklahoma City), but lost badly on the fourth (North Carolina), the only dataset where the raw Gender disparity was not close to statistical noise. This pattern is significant enough to warrant its own discussion (Section 5.3).
- Ethnicity / Race: the most consistently mixed and tool-dependent feature across the benchmark. BiasClean and AIF360 traded wins depending on dataset and metric view, and this feature also produced the two clearest examples of the reference-group-shift / row-overlap dynamic discussed throughout this report.

5.3 Re-examining AIF360's Gender advantage

Considered in isolation, AIF360's record on Gender/Sex (three wins, often by a wide margin) looks like a straightforward strength. The North Carolina result complicates this reading. North Carolina is the only dataset in this benchmark where the raw Gender disparity was not already near the floor of measurable signal, and on that one dataset, BiasClean underperforms AIF360 by a wide margin. The more parsimonious interpretation is that AIF360's apparent Gender advantage elsewhere is driven substantially by the near-null baselines of COMPAS, NIJ, and Oklahoma City's Gender/Sex gaps, rather than a genuine, generalizable strength on this feature category, since both tools have now been shown capable of regressing a near-null gap (AIF360 on Oklahoma City Ethnicity; BiasClean on North Carolina Gender) under what appears to be the same underlying noise-sensitivity mechanism. This report does not claim to have fully resolved this question and recommends it as a priority for any follow-up study (Section 8).

6. Software Correctness Findings During This Benchmark

Two defects in BiasClean's own reporting pipeline were discovered and fixed while producing the results in this report. Both are disclosed here in full, consistent with this project's practice of treating any bug found during independent testing as a required fix, not an inconvenient result to omit.

6.1 Defect: stale per-feature reporting after SVM correction

When BiasClean's optional SVM fairness-enforcement stage runs and produces a result genuinely different from the preceding rebalancing stage, the per-feature numbers in BiasClean's own report and machine-readable audit trail, though not

its single headline composite score, were found to remain frozen at the pre-SVM values. This was traced to the SVM stage's true output being stored in an internal column that the report-generation code did not read from. It was discovered while sanity-checking a BiasClean report ahead of using it in this benchmark, was confirmed present in earlier BiasClean releases as well (not a new regression), and was fixed by making the report-generation code read from the SVM stage's true output whenever it differs from the pre-SVM value.

6.2 Defect: stale exported data after SVM correction

A related, initially separate defect was found immediately afterward: the corrected data file BiasClean exports for external use (the file used as BiasClean's contestant submission in this very benchmark) carried the same stale, pre-SVM values in its named output column, even after the first defect above was fixed. This was fixed by ensuring the exported file's named target column always reflects the pipeline's true final result.

6.3 Scope and impact on this report's results

Both defects only manifest on datasets where the SVM stage runs and produces a result different from ordinary rebalancing. Across this benchmark's five datasets, that condition was met only for Communities & Crime. Neither defect affected that dataset's composite disparity score, which was correct throughout; both affected only the per-feature breakdown and the exported file used for this benchmark's own comparison. The Communities & Crime results reported in Section 4.2 were verified, after both fixes, to be unaffected by either defect.

7. Strengths, Limitations, and When to Use Each Tool

7.1 BiasClean

Strengths	Limitations
• Strongest performance on high-cardinality protected features (many-category Region and Age fields), where it won every judged comparison in this benchmark. • Combined, weight-prioritized multi-feature correction in a single pass, with no equivalent in AIF360's native workflow. • A disclosed, statistically-motivated action threshold that avoided a real regression on a near-null-signal dataset (Oklahoma City) that AIF360 did not avoid. • Full audit trail and machine-readable output designed for non-technical, auto-detected use.	• Its action threshold is a single fixed value (0.05) that does not currently adapt to sample size or group count, the same numeric gap can be treated differently on a small dataset than a large one without the threshold reflecting that. • Demonstrated capacity to widen an already-near-null gap on at least one feature (North Carolina, Gender), the same class of risk this report attributes to AIF360 on Oklahoma City, evidence that neither tool is immune to it. • Correcting one feature can measurably shift another feature's group composition as a side effect (the row-overlap dynamic observed on COMPAS and North Carolina Ethnicity).

7.2 AIF360 (Reweighting, used natively)

Strengths	Limitations
• Strong, consistent performance on Gender/Sex across three of four datasets tested, though see Section 5.3 for a caveat on how much of this reflects genuine skill versus near-null baselines. • Corrects unconditionally, so it will always attempt a correction where BiasClean's threshold declines to act, useful when a practitioner specifically wants every measured disparity addressed regardless of magnitude. • Well-established, independently maintained, and widely used outside this benchmark, with broad community and academic scrutiny.	• Consistently the weaker tool on high-cardinality features across every dataset tested, likely reflecting Reweighting's binary (privileged-versus-unprivileged) design. • No action threshold: in this benchmark, this directly caused a measurable regression on a near-null-signal feature (Oklahoma City, Ethnicity) that a threshold-based tool avoided. • No native multi-feature combined correction, produces one independently-corrected file per protected feature rather than a single coherent corrected dataset.

7.3 Practical guidance

Based on the evidence in this report, the following is offered as practical, evidence-based guidance rather than a categorical recommendation for either tool in all circumstances:

- Prefer BiasClean when the protected feature of interest has many categories (region, detailed age bands, or similarly high-cardinality attributes), or when a single combined correction across multiple protected features in one pass is required.
- Prefer BiasClean when the dataset's disparities are plausibly close to the noise floor and an unconditional correction risks manufacturing an effect that is not reliably present, particularly for smaller datasets or smaller subgroups within a larger dataset.
- Prefer AIF360 when a binary protected attribute (a two-category Gender/Sex field, for example) is the primary concern and the practitioner wants any measured disparity addressed regardless of its statistical reliability.
- In either case, this benchmark's central methodological finding, that both tools can regress an already-small gap under the right conditions, argues for verifying any single correction against an independent measure (as this report does with Aequitas) rather than trusting either tool's self-reported improvement figures in isolation.

8. Limitations of This Study

- This benchmark evaluates one specific configuration of each tool: BiasClean's default weight-prioritized reweighing (with SVM enforcement where triggered), and AIF360's default Reweighing preprocessing algorithm. Neither tool's full range of available algorithms was tested.
- All five datasets are drawn from the criminal-justice domain in the United States (except BiasClean's domain-weighting framework, which is UK-calibrated); findings may not generalize to other domains (healthcare, finance, hiring) or other jurisdictions' legal and demographic contexts.
- Whether BiasClean's fixed 0.05 action threshold should adapt to sample size and group count, rather than remaining fixed, is identified in Section 3.2 as an open design question not resolved by this study.
- Section 5.3's finding regarding AIF360's Gender/Sex results is offered as a plausible interpretation of the available evidence, not a fully isolated causal claim; a dedicated follow-up study varying baseline gap size while holding other factors constant would be needed to confirm it.
- Aequitas's own default action threshold ($\tau = 0.8$) was not used in this benchmark's scoring; a continuous disparity-ratio measure was used instead specifically to avoid importing a second, differently motivated threshold into the judging process itself.

9. Conclusion

Under blind, independent adjudication by a third-party tool, BiasClean outperformed AIF360's native Reweighing algorithm on the majority of dataset–feature comparisons in this benchmark, with its clearest advantage on high-cardinality protected features and its most notable risk, shared with AIF360, on features whose raw disparity is already close to the statistical noise floor. BiasClean's decision to decline correction below a fixed disparity threshold is not merely a self-serving design choice: it is grounded in a body of literature on statistical noise in fairness metrics, is not dissimilar in kind (if different in origin) from a threshold already present in the independent judge used in this very study, and produced a directly observable benefit, avoiding a regression that AIF360, lacking any such threshold, did not avoid.

This report is offered as one input into an ongoing, disclosed process of testing BiasClean against independent tools and independent judgment, including two real defects found and fixed over its course. The authors consider that process itself, and its results being reported in full including unfavorable findings, to be as material to BiasClean's trustworthiness as the comparative results themselves.